

HW0601

11.	Value	rating	distance	d_i	d_i^2
1	12 (7.5)	75 (3)	75 (3)	4.5	20.25
2	7 (4)	165 (7)	165 (7)	-3	9
3	5 (1)	300 (12)	300 (12)	-9	81
4	19 (11)	15 (4)	15 (4)	11	121
5	17 (10)	190 (8)	190 (8)	2	4
6	12 (9.5)	240 (11)	240 (11)	-7.5	56.25
7	9 (6)	120 (5)	120 (5)	2	4
8	19 (11)	60 (2)	60 (2)	4	16
9	3 (4)	270 (10)	270 (10)	-9	81
10	8 (5)	200 (9)	200 (9)	-4	16
11	15 (8)	170 (6.5)	170 (6.5)	7.5	56.25
12	4 (2)	130 (5.5)	130 (5.5)	-7.5	56.25
					<u>454</u>

$$H_0: \rho \geq 0$$

$$H_a: \rho < 0$$

$$a) r_r = 1 - \frac{6}{n(n^2-1)} \sum d_i^2 = 1 - \frac{6 \times 454}{12(12^2-1)} = -0.587$$

$$1) n=12, \alpha=0.05$$

$$r_{12,0.05} = -0.497$$

$$-0.587 < -0.497, \text{ so reject } H_0.$$

12.	d	$\frac{d_i^2}{16}$
d_1	6	49
d_2	7	0.25
d_3	0.5	4
d_4	2	9
d_5	3	1
d_6	-1	25
d_7	-5	20.25
d_8	-4.5	64
d_9	-8	

$$a) r_r = 1 - \frac{6}{n(n^2-1)} \sum d_i^2 = 1 - \frac{6}{9(9^2-1)} \times 208.5 = -0.7775$$

Conclusion

$$h) H_0: \rho = 0$$

$$H_a: \rho \neq 0 \quad \alpha = 0.05$$

$$r_{9,0.025} = 0.683$$

$$r_r < -0.683, \text{ reject } H_0.$$

QA

n = experiment 0 to 30

$r_{n,\alpha/2}$ is the critical value, so for $r_{n,\alpha/2} = 0.683$, this is the critical value, in the $\alpha/2$ of -0.683